Task 4.5 – Perform Comparative Analysis

4.5.1 Compare All Solution Versions

Objective

The objective of this analysis was to evaluate the performance of three customer support architectures:

- **Baseline LLM**
- **Solution V1 – Naive RAG**
- **Solution V2 – Hybrid RAG (Fine-Tuned Intent Router + Retrieval)**

All three solutions were evaluated on the same held-out test dataset using SOP-grounded references to ensure a fair comparison.

Comparative Results

Metric	Baseline	Naive RAG	Hybrid RAG
ROUGE-1	0.0712	0.0836	0.7441
ROUGE-L	0.0427	0.0697	0.7251
BLEU	0.000002	0.000020	0.5904

Quality Evaluation

Metric	Baseline	Naive RAG	Hybrid RAG
Format Adherence	Lower	Improved	100%

Metric	Baseline	Naive RAG	Hybrid RAG
Intent Accuracy	Low	Moderate	High
Output Consistency	Lower	High	High
Hallucination Frequency	High	Reduced	Lowest

Comparative Analysis

Baseline LLM

The baseline model generated responses without access to company policies or domain-specific knowledge. As a result, responses frequently lacked policy accuracy and occasionally contained hallucinated information. The lowest ROUGE and BLEU scores confirmed limited similarity with the reference responses.

Solution V1 – Naive RAG

Naive RAG improved the overall response quality by retrieving relevant SOP documents before generation. The retrieved context reduced hallucinations and improved lexical similarity compared with the baseline model. However, retrieval depended directly on the user's query, making it less effective for ambiguous or incomplete customer requests.

Solution V2 – Hybrid RAG

Hybrid RAG achieved the best overall performance by combining parameter-efficient fine-tuning with retrieval.

Instead of retrieving documents directly from the customer query, the fine-tuned intent router first predicted the customer's intent and support category in structured JSON format. The predicted intent was then mapped to an optimized retrieval query before performing vector similarity search.

This additional routing step significantly improved retrieval quality, resulting in:

- higher ROUGE scores,

- substantially improved BLEU score,
- better policy grounding,
- lower hallucination frequency,
- and more consistent responses.

The evaluation demonstrated that combining fine-tuning with retrieval produced the most accurate and reliable customer support responses.

4.5.2 Findings and Recommendations

Key Findings

- Fine-tuning significantly improved intent recognition and retrieval accuracy.
- Hybrid RAG consistently outperformed both the baseline model and Naive RAG across all evaluation metrics.
- Using predicted intent instead of raw customer queries improved SOP retrieval quality.
- Structured JSON outputs achieved 100% format adherence throughout the evaluation.
- Hallucinations were substantially reduced after introducing policy-grounded retrieval.

Limitations

- The project focused on a single customer support domain (Refund Support).
- Only Top-1 document retrieval (`k = 1`) was evaluated.
- Multi-turn conversations and conversational memory were outside the project scope.
- Evaluation was performed on a relatively small held-out dataset compared with production-scale customer support systems.

Recommendations

- Expand the system to support multiple customer support domains such as shipping, billing, technical support, and account management.
- Introduce Hybrid Search (keyword + vector search) to improve retrieval for ambiguous queries.
- Apply document reranking to further improve retrieval precision.
- Evaluate larger instruction-tuned models where computational resources permit.

Future Work

Future enhancements may include:

- Multi-agent customer support workflows.
- Real-time enterprise API integration.
- Dynamic knowledge base updates.
- Human-in-the-loop verification for critical responses.
- Multilingual customer support.
- Continuous fine-tuning using customer feedback.
- Deployment as a scalable enterprise customer support platform.

Conclusion

The experimental results demonstrate that **Hybrid RAG successfully combines the strengths of parameter-efficient fine-tuning and retrieval-augmented generation**. Compared with both the baseline model and Naive RAG, the proposed approach produced more accurate intent classification, improved retrieval quality, significantly reduced hallucinations, and generated policy-compliant customer support responses. These findings indicate that Hybrid RAG is a practical and scalable solution for enterprise customer support automation.

Project Repository

GitHub:

https://github.com/ssprajapati2021/Hybrid-RAG-Fine-Tuning
